

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
%fit_data_file = 'FitEn_cut800br200_2D_fit_bgSer.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
%fit_data_file = 'FitEn_cut800br100_2D_dirGH_dE20_fix_bg_slope.mat'
fit_data_file = 'FitEn_cut800ref110_2D_dirGH_dE20_fix_bg_slope.mat'

```

```

fit_data_file =
'FitEn_cut800ref110_2D_dirGH_dE20_fix_bg_slope.mat'

```

```

if ~exist('fit_resEi800','var')
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi800')
        fit_resEi800 = ld.fit_resEi800;
    else
        fit_resEi800 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_cut_data_file = 'multicuts_fit_dataGH_ei800meV.mat';
if ~exist('cuts2fit800','var')
    ld = load(fullfile(data_path,fit_cut_data_file));
    cuts2fit800 = ld.cuts2fit800;
end
cuts2fit = cuts2fit800;
fit_res = fit_resEi800;
mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
c1 = {cuts2fit800.cut2fit010_dir100}

```

```

c1 = 1x1 cell array
{1x1 sqw}

```

```

for i=1:numel(c1)
    c1{i} = c1{i}.set_instrument(mi);
    c1{i} = c1{i}.set_sample(sample);

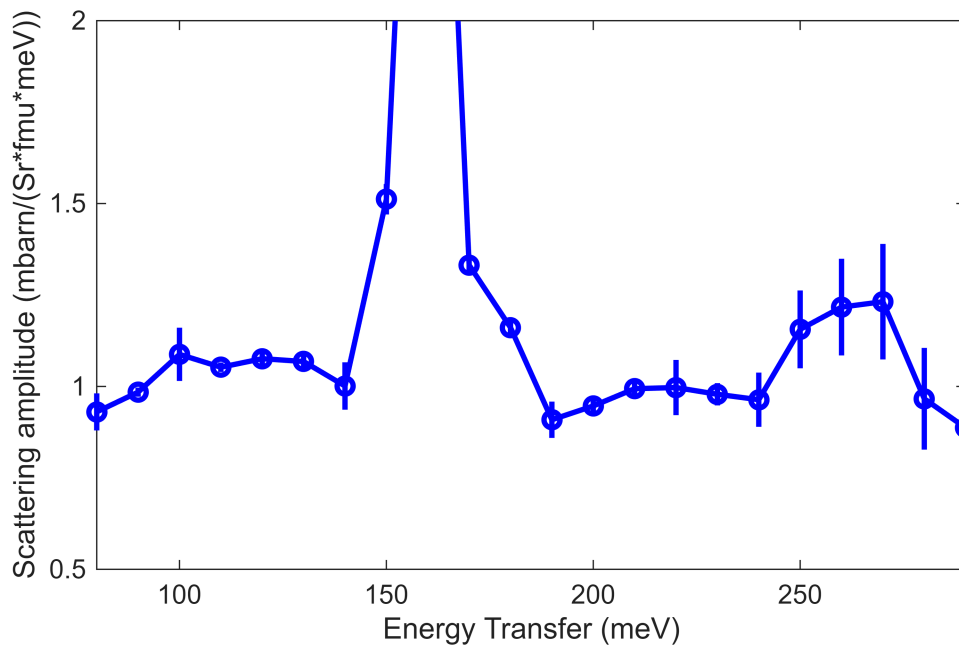
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
cuts2fit800.cut2fit010_dir100 = c1{1};
the2Dcuts =c1;

```

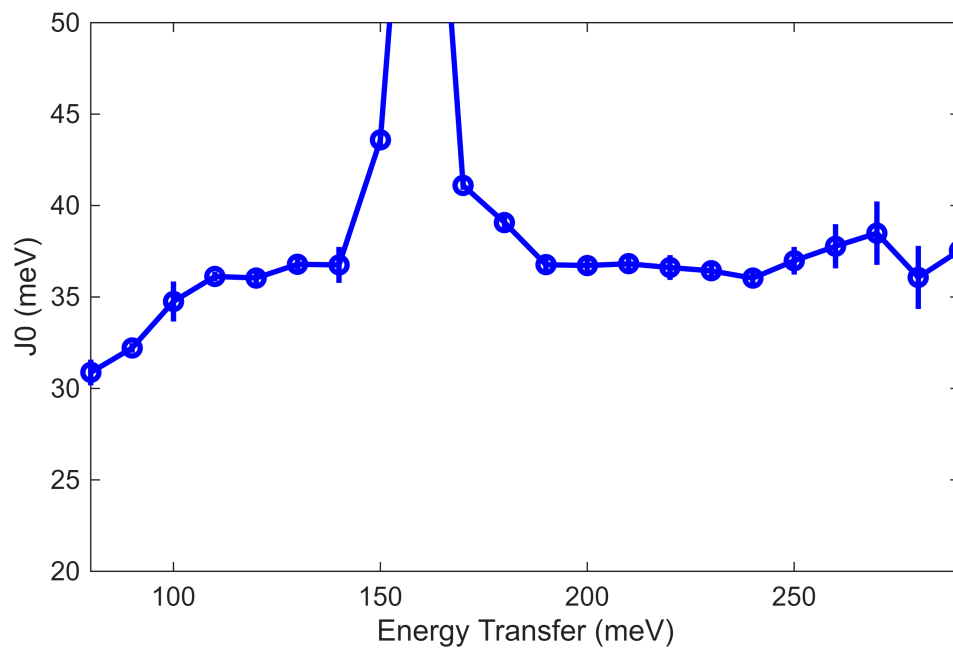
```

acolor b
Sgn1 = fit_res.S;
pd(Sgn1);ly 0.5 2;liny;keep_figure

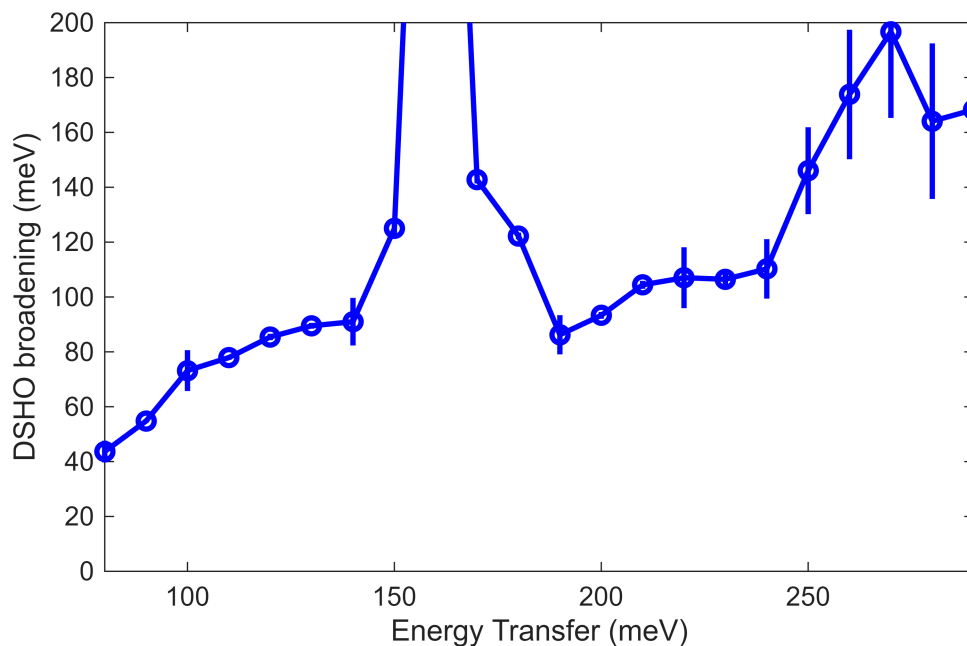
```



```
J0 = fit_res.J0;
pd(J0);ly 20 50;keep_figure
```



```
% incorrect units fixture
G = fit_res.gamma;
G.s_axis.units = 'meV';
fit_res.gamma = G;
pd(G);ly 0 200;keep_figure
```



```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1));
e_min = Sgn1.x(1);
```

```
En = 160
```

```
En =
160
```

```
dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10;
en_idx = floor((En-e_min)/dE)+1;
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);
the2Dcuts =c1;

%init_bg_params =
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg
%in case of 2-D fitting
%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
%the2Dcuts{3} = cut(the2Dcuts{3},[-1,0.02,1],[]);
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);
```

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33167

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 13451 pixels -- processing data... -----> included 13081 pixels

Number of points in input object: 203609

Fraction of object selected: 6.61% (= 13451 points) ; Access speed: 0.15GB/sec

```

Fraction of object retained:      6.42% (=      13081 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  21.3%; Transf. time frac.:  70.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    9655 pixels -- processing data... -----> retained      9281 pixels
-----
Number of points in input object: 203664
    Fraction of object selected:   4.74% (=      9655 points) ; Access speed:      0.46GB/sec
    Fraction of object retained:   4.56% (=      9281 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  13.8%; Transf. time frac.:  49.0%; Resulting pix preparation:  23.8%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   58965 pixels -- processing data... -----> included     58231 pixels
-----
Number of points in input object: 203609
    Fraction of object selected:  28.96% (=     58965 points) ; Access speed:      1.79GB/sec
    Fraction of object retained:  28.60% (=     58231 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  27.5%; Transf. time frac.:  66.1%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   32453 pixels -- processing data... -----> retained     31166 pixels
-----
Number of points in input object: 203664
    Fraction of object selected:   15.93% (=     32453 points) ; Access speed:      1.66GB/sec
    Fraction of object retained:   15.30% (=     31166 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  16.0%; Transf. time frac.:  62.4%; Resulting pix preparation:  21.5%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   98780 pixels -- processing data... -----> included     97483 pixels
-----
Number of points in input object: 203609
    Fraction of object selected:   48.51% (=     98780 points) ; Access speed:      2.05GB/sec
    Fraction of object retained:   47.88% (=     97483 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  23.6%; Transf. time frac.:  72.8%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    81627 pixels -- processing data... -----> retained     79994 pixels
-----
Number of points in input object: 203664
    Fraction of object selected:   40.08% (=     81627 points) ; Access speed:      2.07GB/sec
    Fraction of object retained:   39.28% (=     79994 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  16.8%; Transf. time frac.:  64.1%; Resulting pix preparation:  19.0%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   39296 pixels -- processing data... -----> included     38258 pixels
-----
Number of points in input object: 203609
    Fraction of object selected:   19.30% (=     39296 points) ; Access speed:      1.74GB/sec
    Fraction of object retained:   18.79% (=     38258 points)

**** Total time in cut(sqw)   :      0.0sec Including:
    Data access time fraction:  17.8%; Transf. time frac.:  76.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    21368 pixels -- processing data... -----> retained     20436 pixels
-----

```


Number of points in input object: 203664

Fraction of object selected: 10.49% (= 21368 points) ; Access speed: 1.48GB/sec

Fraction of object retained: 10.03% (= 20436 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 13.8%; Transf. time frac.: 62.5%; Resulting pix preparation: 23.5%

Warning: Dataset 1:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.2821

Beginning fit (max 20 iterations)

Starting point

Total time = 1.2709s Reduced Chi^2 = 1.855

Free parameter values:

462.2	3.81	72.35	0.3467	0.3467	0.3467	0.3467
-------	------	-------	--------	--------	--------	--------

Iteration = 1

Total time = 5.9994s Reduced Chi^2 = 1.3606 Levenberg-Marquardt = 0.1

Free parameter values:

438.4	3.072	66.78	0.6456	0.6997	0.713	0.7339
-------	-------	-------	--------	--------	-------	--------

Iteration = 2

Total time = 10.716s Reduced Chi^2 = 1.4623 Levenberg-Marquardt = 0.01

Free parameter values:

165.2	1.688	47.49	0.7646	0.9011	0.9005	0.8629
-------	-------	-------	--------	--------	--------	--------

Total time = 11.9784s Reduced Chi^2 = 1.3202 Levenberg-Marquardt = 0.1

Free parameter values:

358	2.597	60.85	0.7152	0.8139	0.8202	0.8105
-----	-------	-------	--------	--------	--------	--------

Iteration = 3

Total time = 16.815s Reduced Chi^2 = 1.4445 Levenberg-Marquardt = 0.01

Free parameter values:

131.1	1.46	43.56	0.7907	0.9439	0.9403	0.8897
-------	------	-------	--------	--------	--------	--------

Total time = 18.0556s Reduced Chi^2 = 1.3114 Levenberg-Marquardt = 0.1

Free parameter values:

286.8	2.224	55.26	0.747	0.8665	0.8697	0.8438
-------	-------	-------	-------	--------	--------	--------

Iteration = 4

Total time = 22.8326s Reduced Chi^2 = 1.3097 Levenberg-Marquardt = 0.01

Free parameter values:

115.5	1.279	41.15	0.8212	0.9975	0.9896	0.9221
-------	-------	-------	--------	--------	--------	--------

Iteration = 5

Total time = 27.5151s Reduced Chi^2 = 1.2259 Levenberg-Marquardt = 0.001

Free parameter values:

76.07	0.787	36.04	0.9847	1.293	1.248	1.094
-------	-------	-------	--------	-------	-------	-------

Iteration = 6

Total time = 32.2059s Reduced Chi^2 = 1.1844 Levenberg-Marquardt = 0.0001

Free parameter values:

79.24	0.8466	36.49	0.9852	1.299	1.249	1.096
-------	--------	-------	--------	-------	-------	-------

Iteration = 7

Total time = 36.8573s Reduced Chi^2 = 1.1836 Levenberg-Marquardt = 1e-05

Free parameter values:

79.21	0.8469	36.46	0.9869	1.297	1.248	1.096
-------	--------	-------	--------	-------	-------	-------

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```
-----
1          1
2          8
3      79.21 +/- 8.02
4      0.8469 +/- 0.06603
5          0
6      36.46 +/- 0.7599
7          0
8          0
9          0
10         0
```

Background parameter values:

Function 1:

```
1      0.9869 +/- 0.05588
2     -0.01577
3     -0.003937
4     -0.000359
```

Function 2:

```
1      1.297 +/- 0.06173
2     -0.01577
3     -0.003937
4     -0.000359
```

Function 3:

```
1      1.248 +/- 0.05081
2     -0.01577
3     -0.003937
4     -0.000359
```

Function 4:

```
1      1.096 +/- 0.04649
2     -0.01577
3     -0.003937
4     -0.000359
```

Covariance matrix for free parameters:

```
-----
1.0000  0.9619  0.9325 -0.4533 -0.7396 -0.7673 -0.5727
0.9619  1.0000  0.9125 -0.5094 -0.8234 -0.8732 -0.6483
0.9325  0.9125  1.0000 -0.4359 -0.6616 -0.7061 -0.5383
-0.4533 -0.5094 -0.4359 1.0000  0.4459  0.4809  0.3553
-0.7396 -0.8234 -0.6616 0.4459  1.0000  0.7849  0.5758
-0.7673 -0.8732 -0.7061 0.4809  0.7849  1.0000  0.6201
-0.5727 -0.6483 -0.5383 0.3553  0.5758  0.6201  1.0000
```

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 203664 pixels -- processing data... -----> included 203664 pixels

Number of points in input object: 203664

```
Fraction of object selected: 100.00% (= 203664 points) ; Access speed: 1.93GB/sec
Fraction of object retained: 100.00% (= 203664 points)
```

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 44.8%; Transf. time frac.: 44.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 9264 pixels -- processing data... -----> included 9264 pixels

Number of points in input object: 9264

```
Fraction of object selected: 100.00% (= 9264 points) ; Access speed: 1.15GB/sec
```

Fraction of object retained: 100.00% (= 9264 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 9.3%; Transf. time frac.: 81.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 31166 pixels -- processing data... -----> included 31166 pixels

Number of points in input object: 31166
Fraction of object selected: 100.00% (= 31166 points) ; Access speed: 1.66GB/sec
Fraction of object retained: 100.00% (= 31166 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 15.6%; Transf. time frac.: 77.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 79994 pixels -- processing data... -----> included 79994 pixels

Number of points in input object: 79994
Fraction of object selected: 100.00% (= 79994 points) ; Access speed: 1.02GB/sec
Fraction of object retained: 100.00% (= 79994 points)

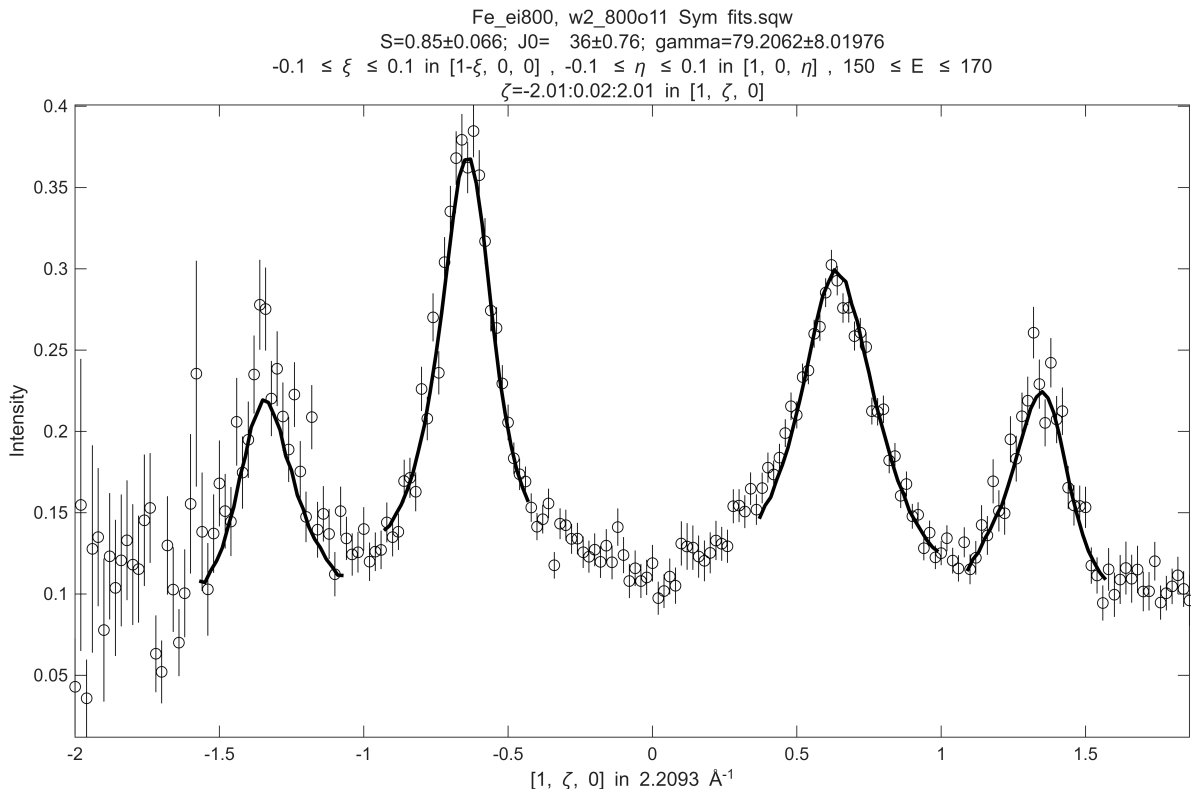
**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 22.4%; Transf. time frac.: 74.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 20436 pixels -- processing data... -----> included 20436 pixels

Number of points in input object: 20436
Fraction of object selected: 100.00% (= 20436 points) ; Access speed: 1.29GB/sec
Fraction of object retained: 100.00% (= 20436 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 14.5%; Transf. time frac.: 78.8%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step



1 of 1; Read data for 203664 pixels -- processing data... -----> retained 203664 pixels

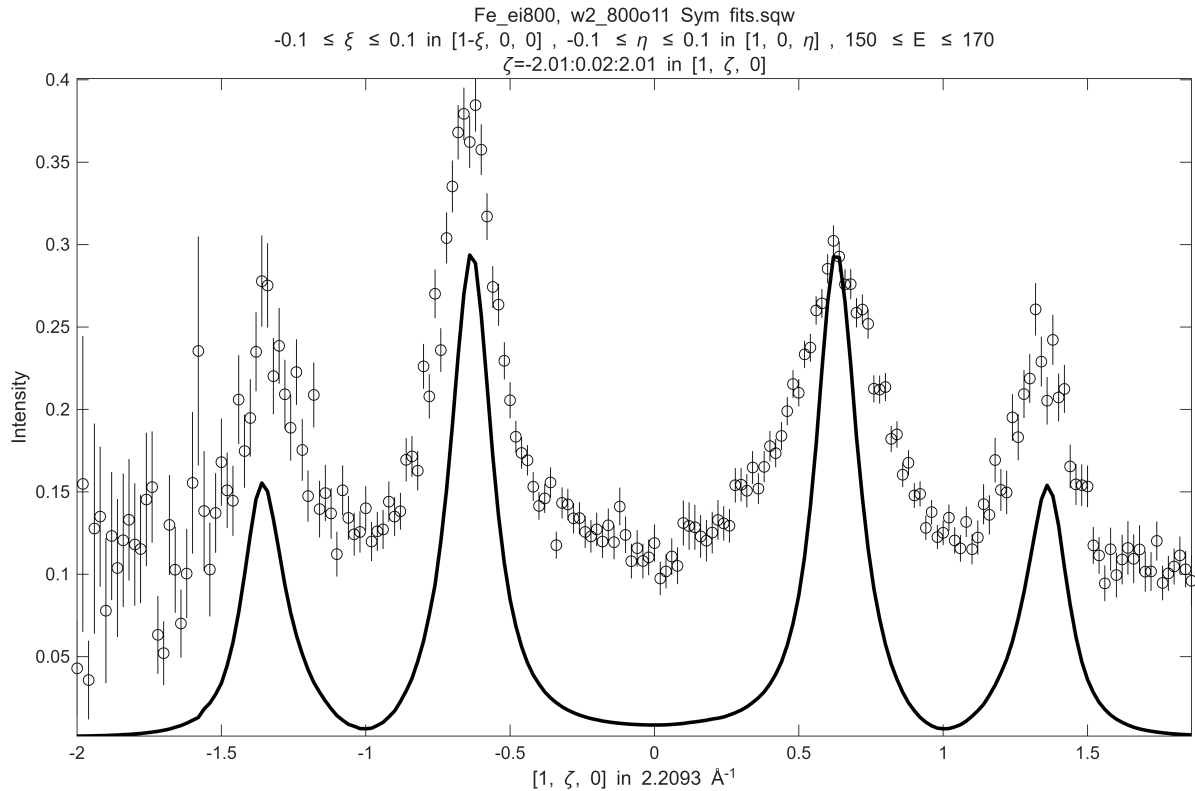
Number of points in input object: 203664

Fraction of object selected: 100.00% (= 203664 points) ; Access speed: 1.80GB/sec

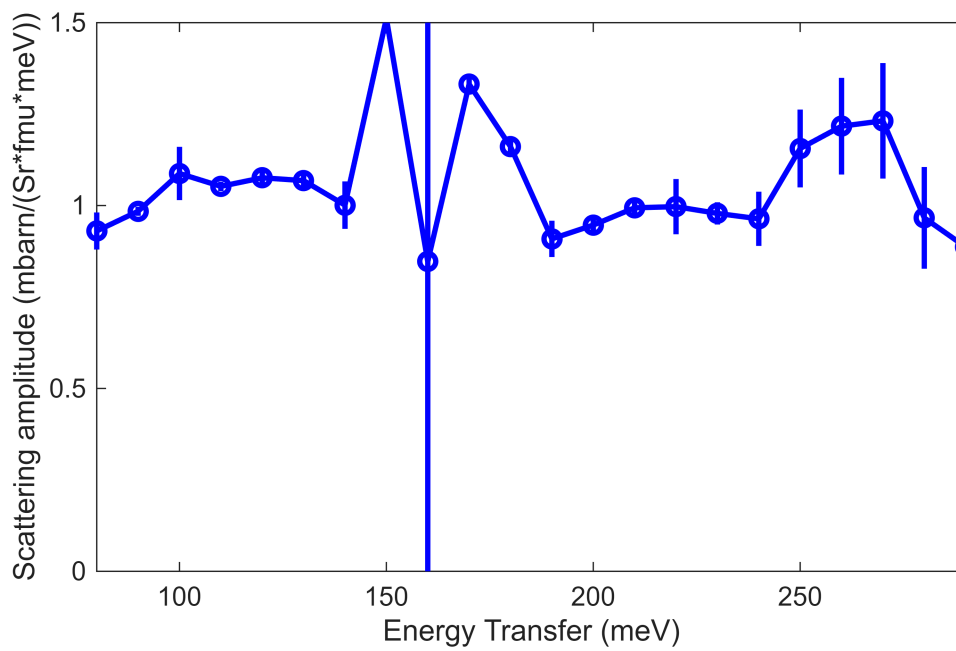
Fraction of object retained: 100.00% (= 203664 points)

**** Total time in cut(sqw) : 0.0sec Including:

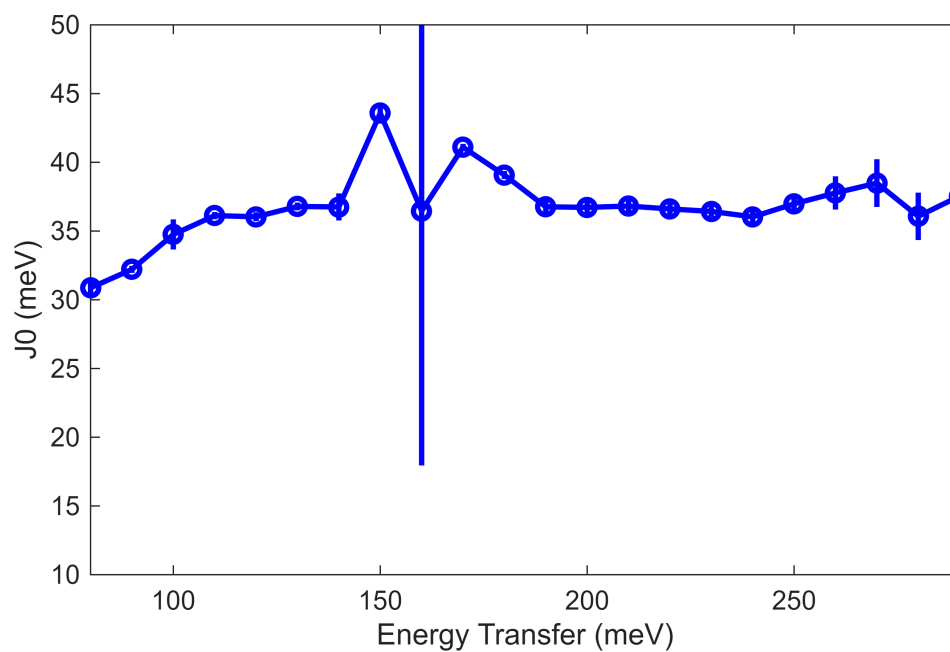
Data access time fraction: 18.7%; Transf. time frac.: 51.0%; Resulting pix preparation: 29.0%



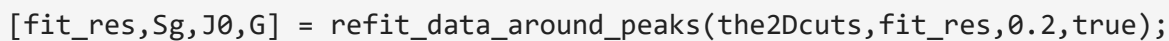
```
%fit_res.all_fit_par(en_idx) = fit_par;  
%fit_res.all_fit_par(en_idx).p = fit_par.p;  
acolor r  
plot(fit_res.S);ly 0 1.5;liny;  
acolor b  
Sgnl.signal(en_idx) = fit_par.p(4);  
pd(Sgnl);ly 0 1.5;liny;keep_figure
```



```
acolor r
plot(fit_res.J0);ly 10 50;liny;
acolor b
J0.signal(en_idx) = fit_par.p(6);
pd(J0);ly 10 50;keep_figure
```

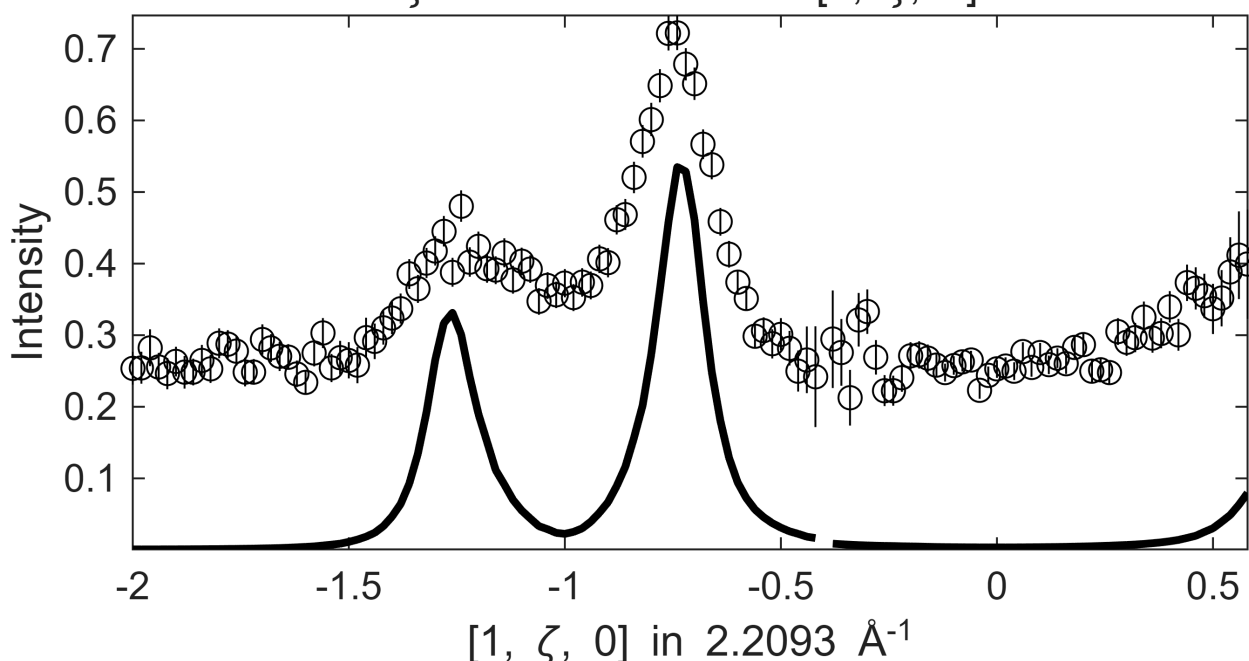


```
acolor r
plot(fit_res.gamma);ly 0 200;liny;
acolor b
G.signal(en_idx) = fit_par.p(3);
pd(G);ly 0 200;keep_figure
```


$$0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 70 \leq E \leq 70 \\ \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


Fe_ei800, w2_800o11 Sym fits.sqw

$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$

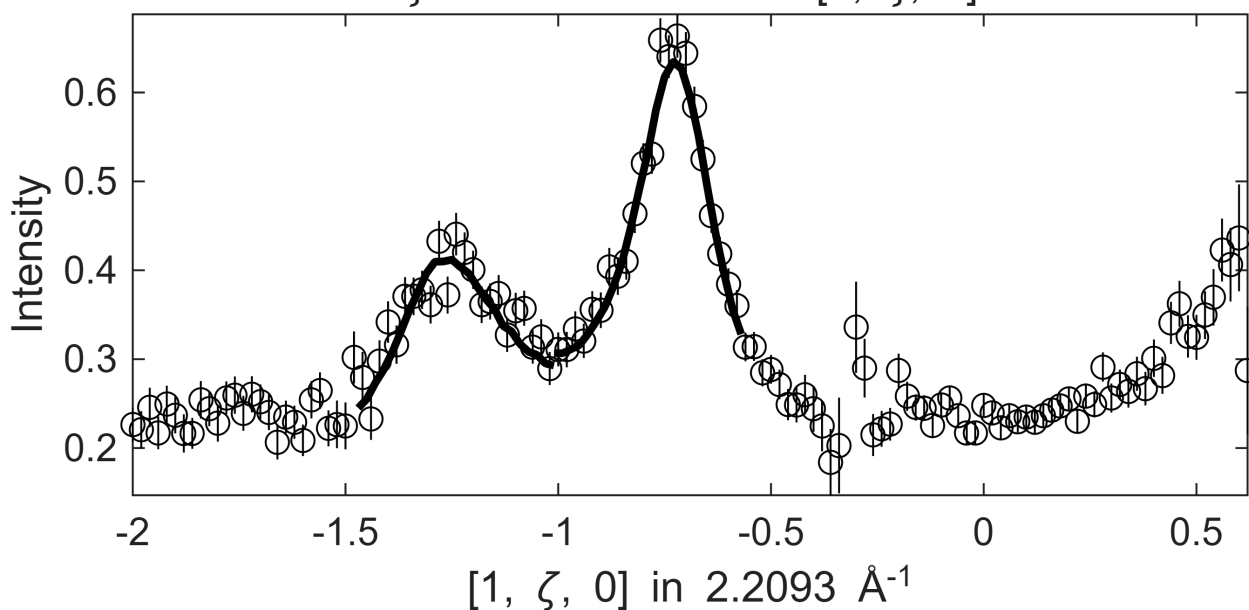


```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 2#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800o11 Sym fits.sqw

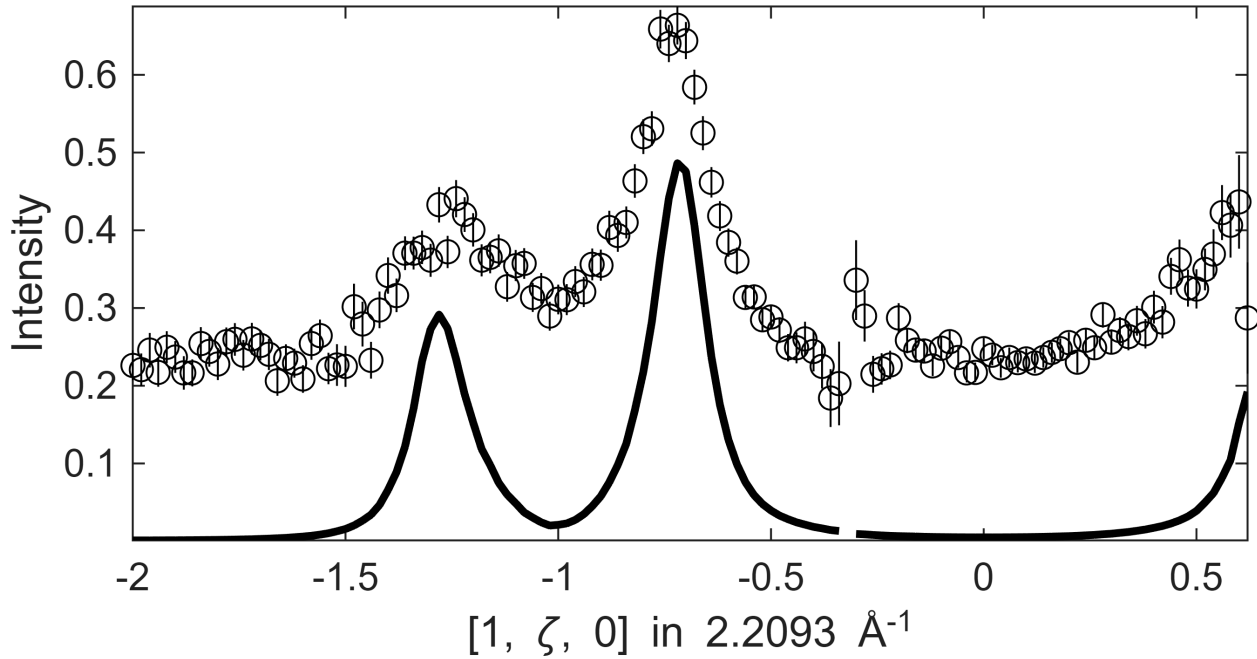
$S=0.97\pm0.036$; $J_0= 32\pm0.29$; $\gamma=52.2348\pm0.974053$

$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei800, w2_800o11 Sym fits.sqw

$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



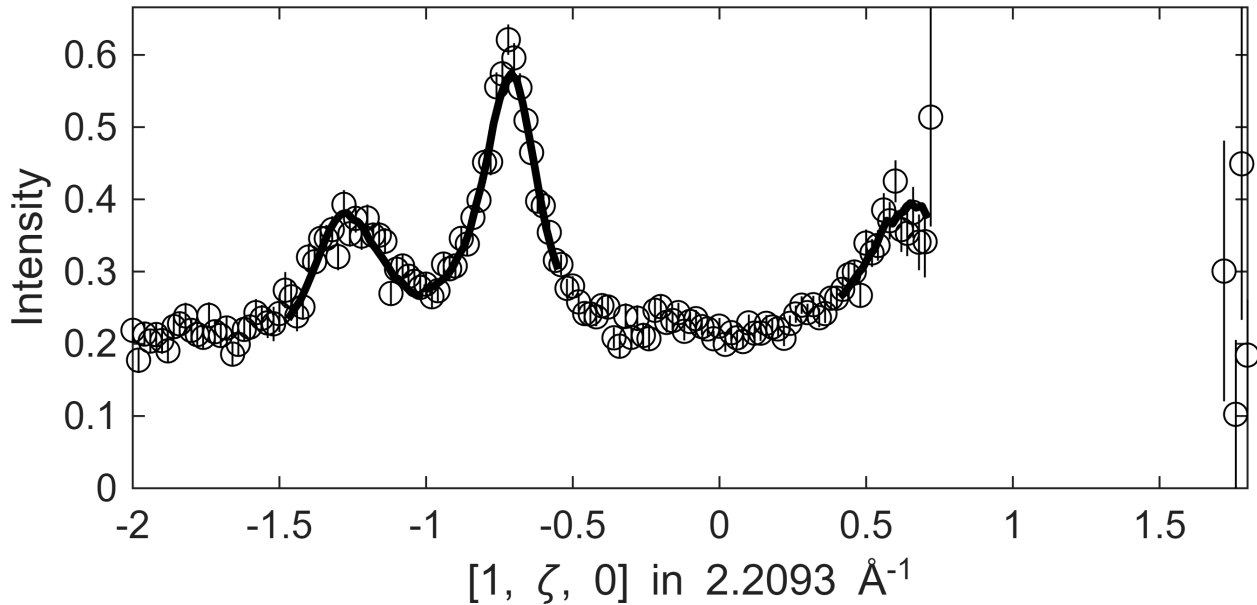
 *** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***

 *** step: 3#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***

Fe_ei800, w2_800o11 Sym fits.sqw

S= 1 ± 0.12 ; J0= 34 ± 1.2 ; gamma= 62.8283 ± 8.86637

$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$$0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 90 \leq E \leq 100 \text{ in } [E, 0, 0], \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$

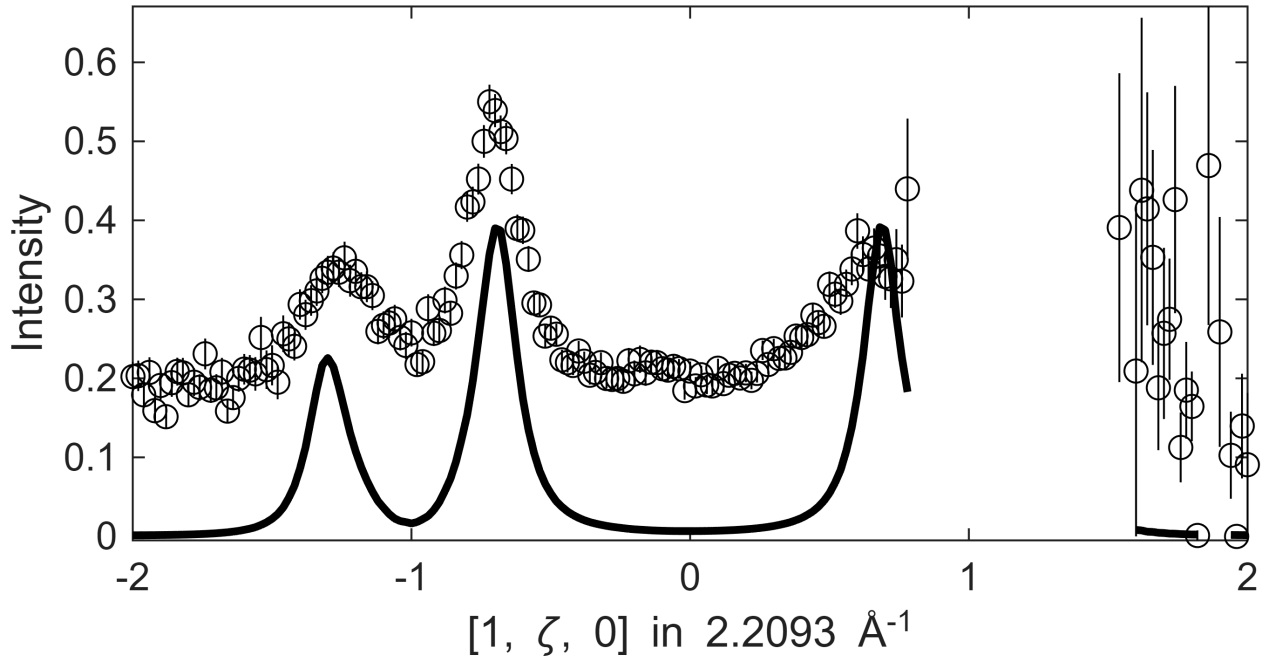

S= 1.1±0.11; J0= 36± 1.2; gamma=73.338±9.17803

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 100 \leq E \leq$$

$$\zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


Fe_ei800, w2_800o11 Sym fits.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



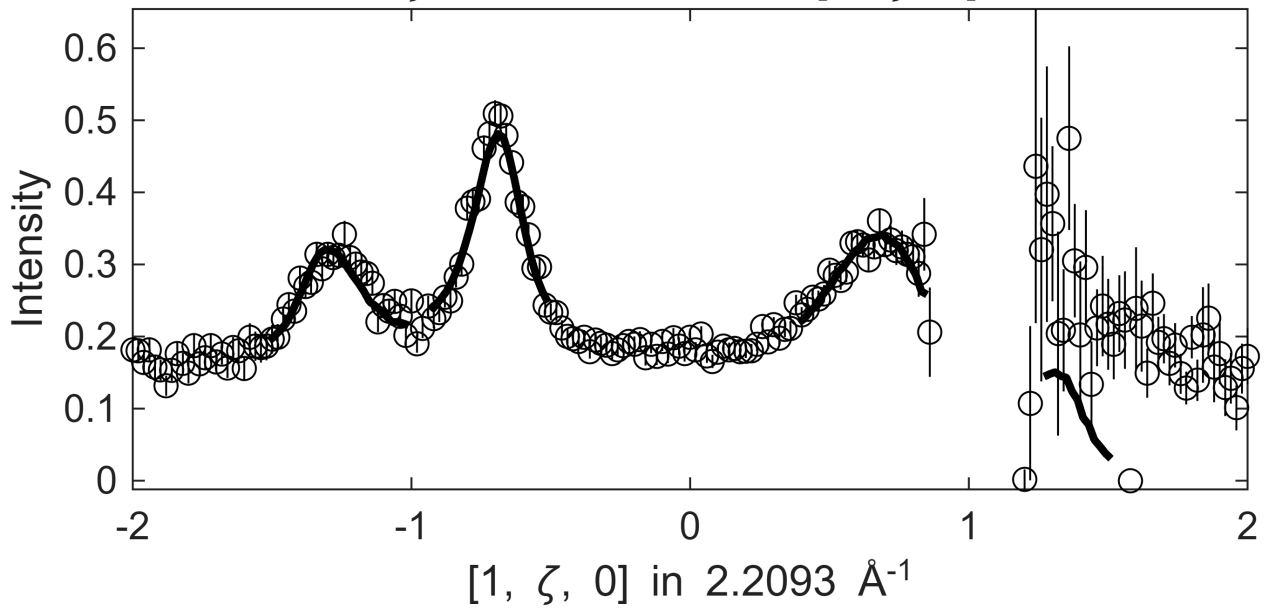
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 5#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:6 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.49751
Warning: Dataset 4:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei800, w2_800o11 Sym fits.sqw

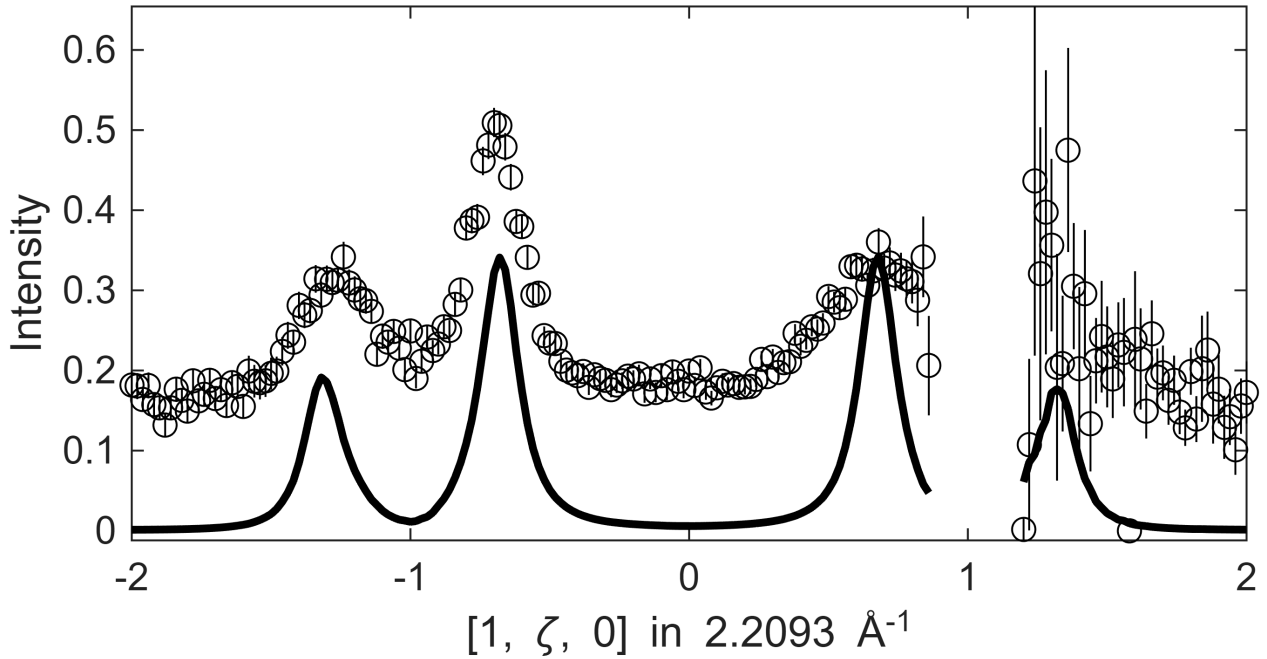
$S=0.86\pm0.11$; $J_0= 35\pm 1.3$; $\gamma=68.2335\pm10.4528$

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei800, w2_800o11 Sym fits.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $110 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



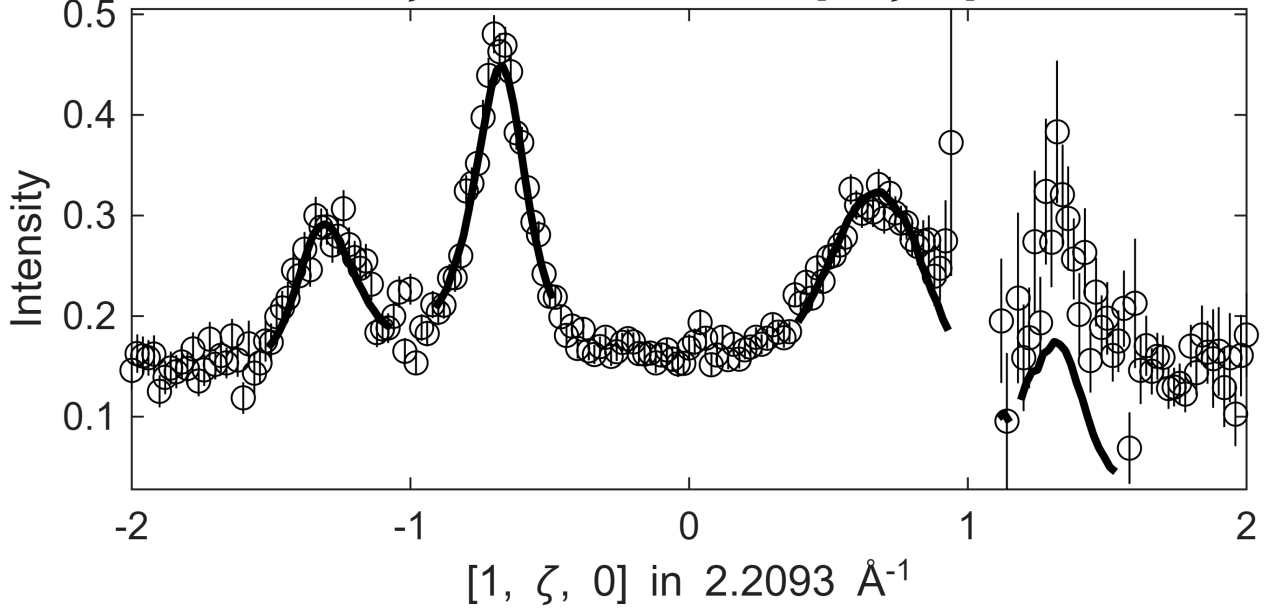
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 6#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:6 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.49751
Warning: Dataset 4:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.5152

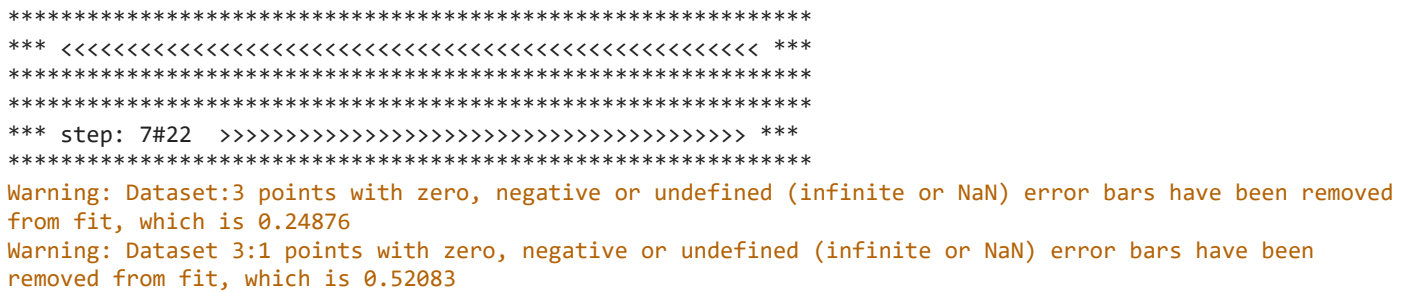
Fe_ei800, w2_800o11 Sym fits.sqw

S=0.98±0.14; J0= 37± 1.5; gamma=81.2198±13.9854

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $120 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$

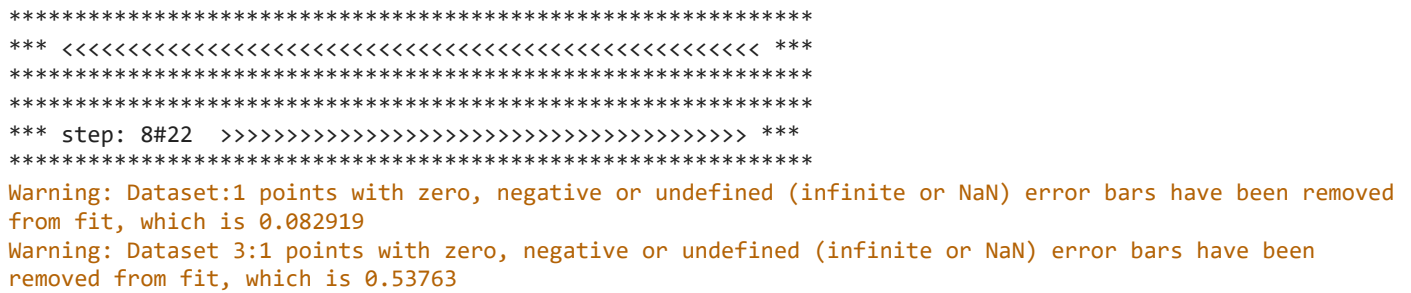


$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $120 \leq E \leq 180$ in $[1, 0, 0]$, $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



S=0.91±0.11; J0= 36± 1; gamma=82.3565±10.0804
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $130 \leq E \leq 170$ in $[1, \zeta, 0]$



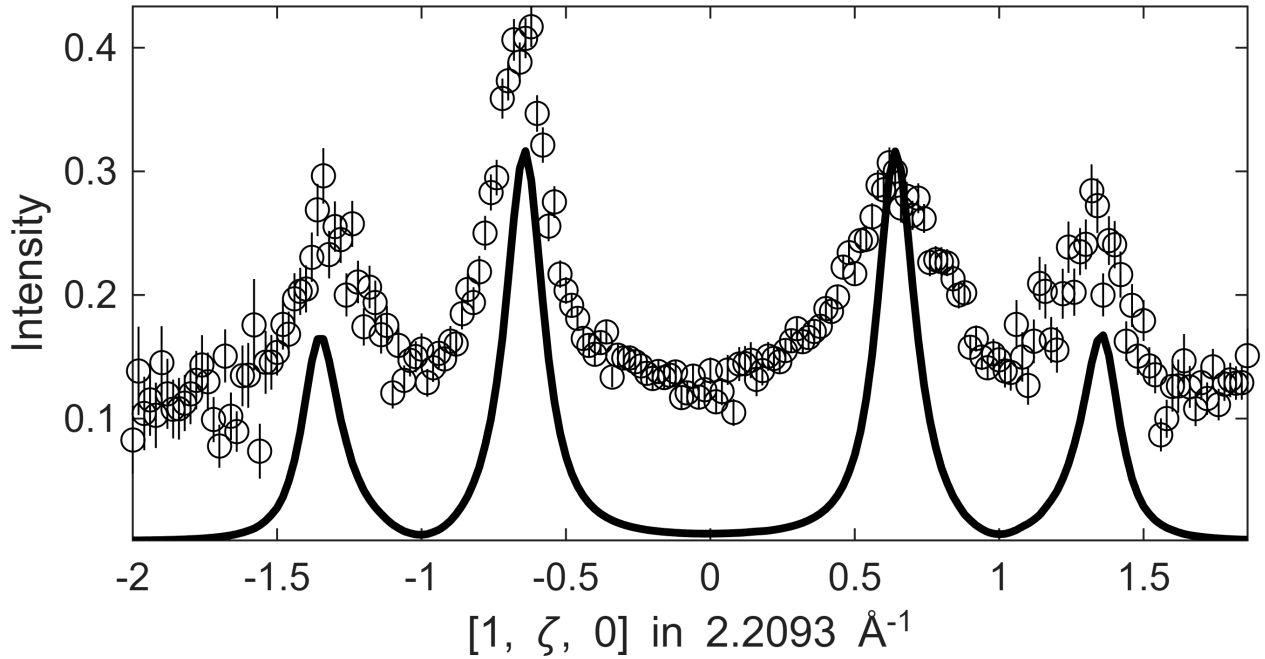
$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 130 \leq E \leq 170 \text{ in } [1, \zeta, 0]$$


S=0.78±0.046; J0= 35±0.41; gamma=67.064±4.12687

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 140 \leq E \leq 160 \text{ in } [1, \zeta, 0]$$


Fe_ei800, w2_800o11 Sym fits.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $140 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



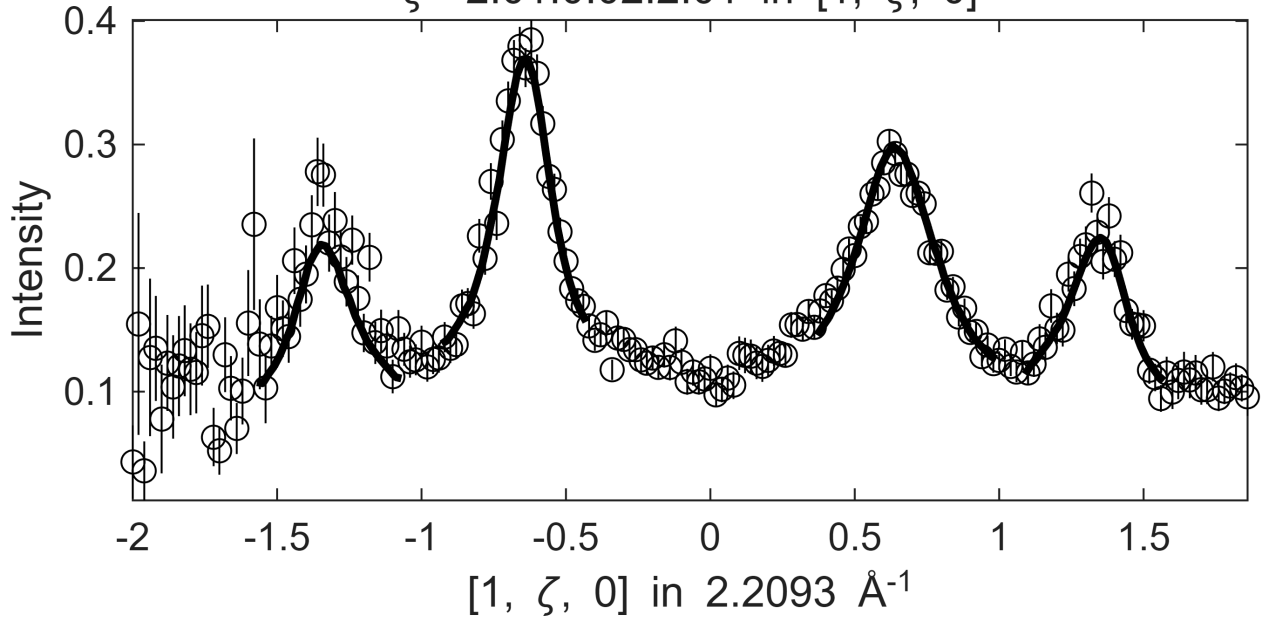
```
*****
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*****
*****
*** step: 9#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.33167
Warning: Dataset 1:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.2821

Fe_ei800, w2_800o11 Sym fits.sqw

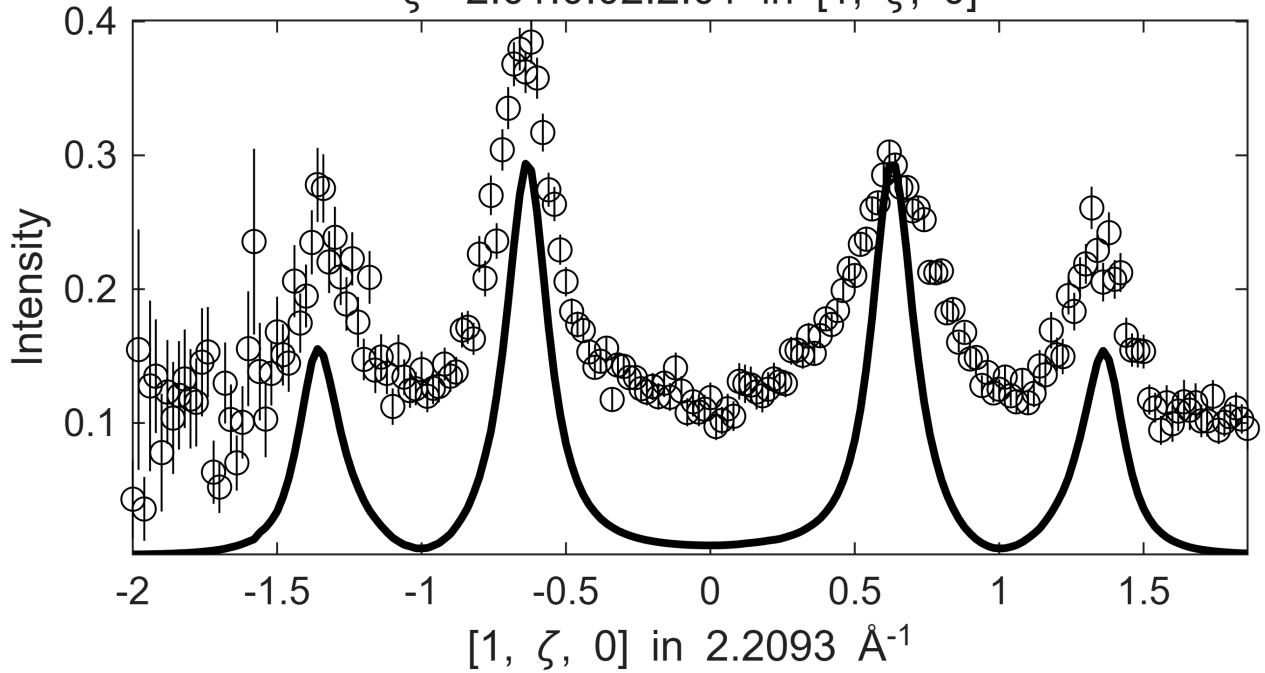
$S=0.85\pm0.066$; $J_0= 36\pm0.76$; $\gamma=79.2505\pm8.0535$

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $150 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei800, w2_800o11 Sym fits.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $150 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



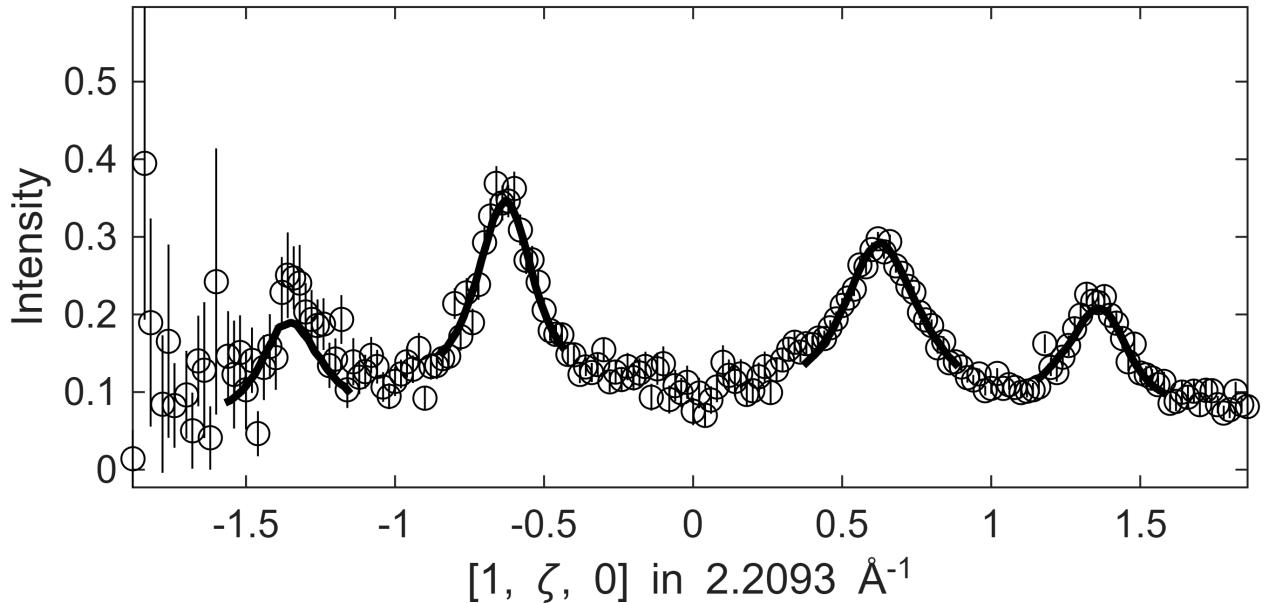
```
*****
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*****
*****
*** step: 10#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.58043
Warning: Dataset 1:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.0303

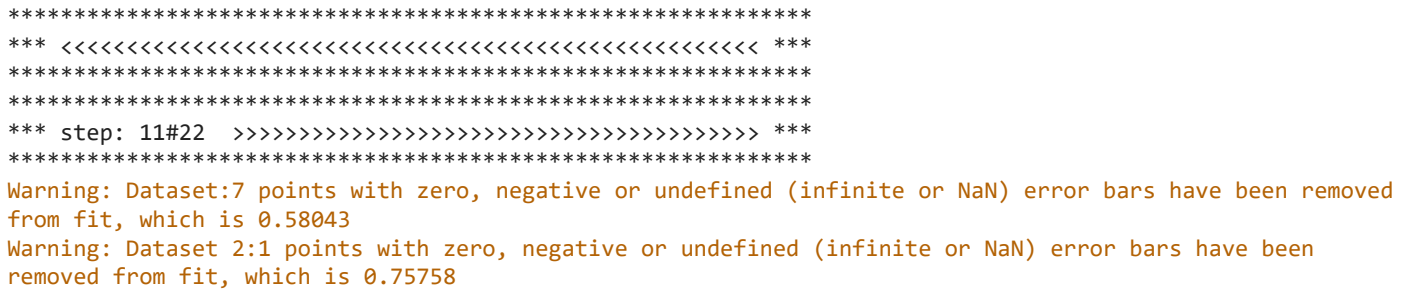
Fe_ei800, w2_800o11 Sym fits.sqw

$S=0.88\pm0.079$; $J_0= 37\pm0.79$; $\gamma=87.4361\pm9.38324$

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $160 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 160 \leq E \leq 200$$

$$\zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


S=0.94±0.097; J0= 37±0.88; gamma=94.6664±11.573
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $170 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $170 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$S=0.61\pm0.068$; $J_0=35\pm0.46$; $\gamma=57.5893\pm7.9114$

$$-0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 180 \leq E \leq 270$$


$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 180 \leq E \leq \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$

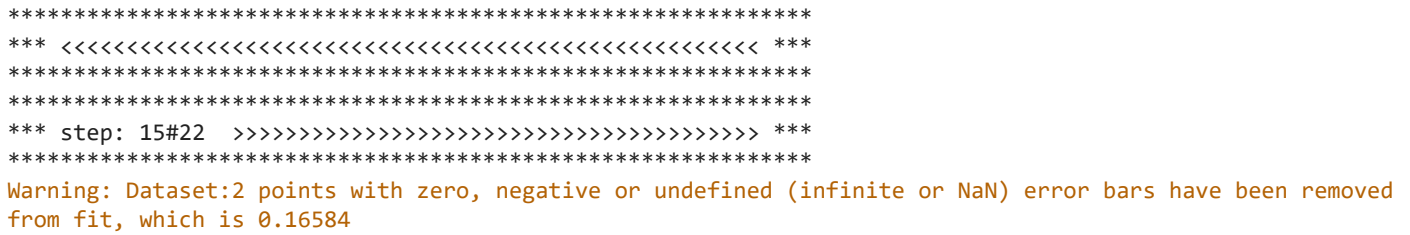

$S=0.62\pm0.062$; $J_0=34\pm0.46$; $\gamma=56.8718\pm4.71461$

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 190 \leq E \leq 210 \text{ in } [190, E, 210],$$


$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 190 \leq E \leq 210 \text{ in } [190, E, 210],$$


S=0.69±0.046; J0= 35±0.43; gamma=71.551±1.90461

$$-0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 200 \leq E \leq 2500 \text{ in } [200, E, 2500], \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 200 \leq E \leq 2000, \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$S=0.65\pm0.11$; $J_0=35\pm0.58$; $\gamma=69.4928\pm12.8065$

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $210 \leq E \leq 290$ in $[E, 0, 0]$, $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 210 \leq E \leq 215, \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$S=0.63\pm0.031$; $J_0=35\pm0.36$; $\gamma=70.0721\pm1.59222$

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 220 \leq E \leq 240 \text{ in } [220, E, 240], \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $220 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



S=0.62±0.098; J0= 34±0.47; gamma=73.2203±12.2342

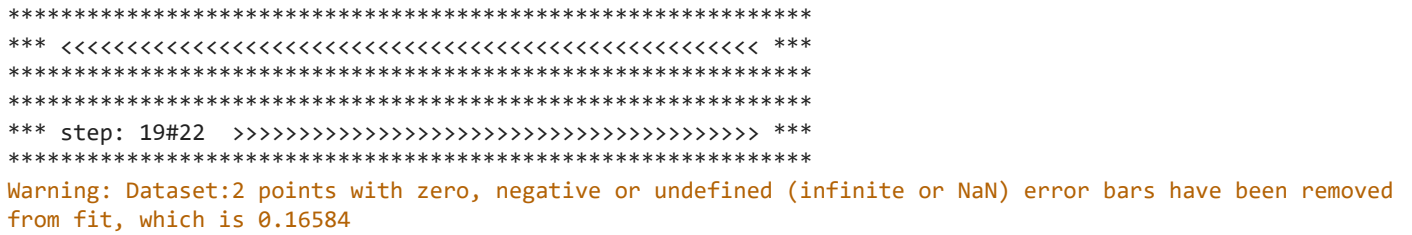
$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 230 \leq E \leq 270 \text{ in } [1, \zeta, 0]$$


$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 230 \leq E \leq 270$$

$$\zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$S=0.71\pm0.057$; $J_0=35\pm0.36$; $\gamma=93.7169\pm4.75796$

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 240 \leq E \leq 260 \text{ in } [240, E, 260], \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$

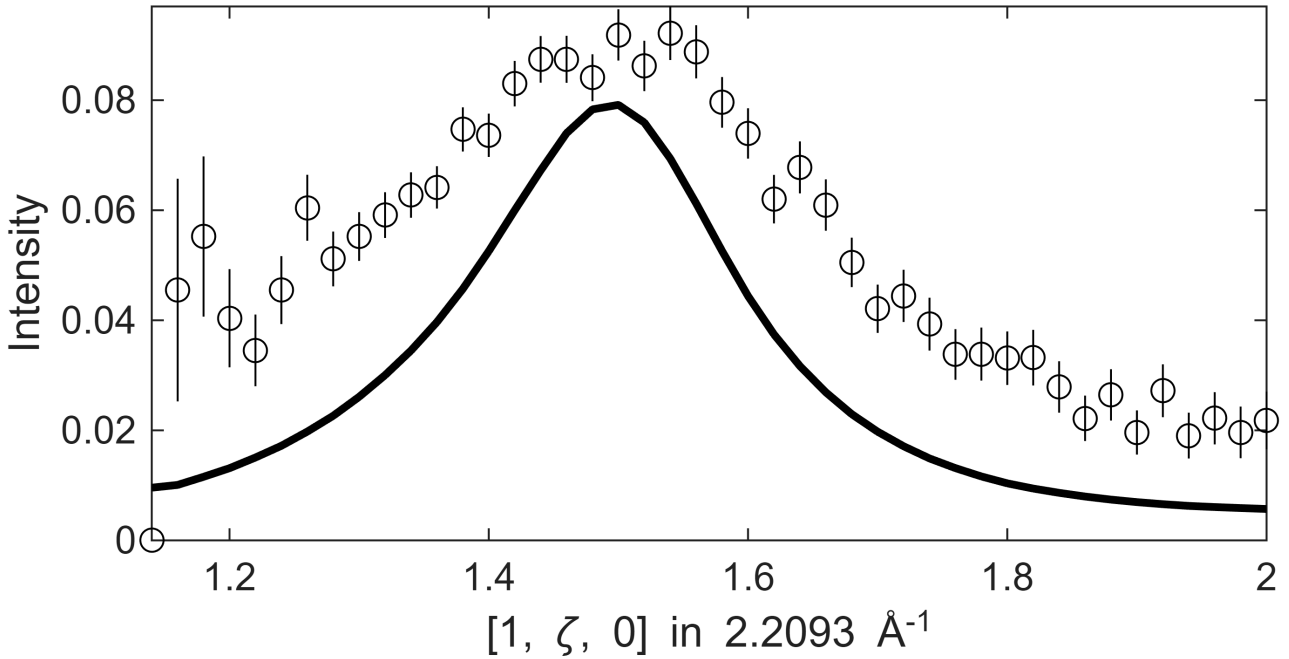

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 240 \leq E \leq 250$$


S=0.74±0.097; J0= 35±0.44; gamma=115.87±10.1771
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $250 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei800, w2_800o11 Sym fits.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $250 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



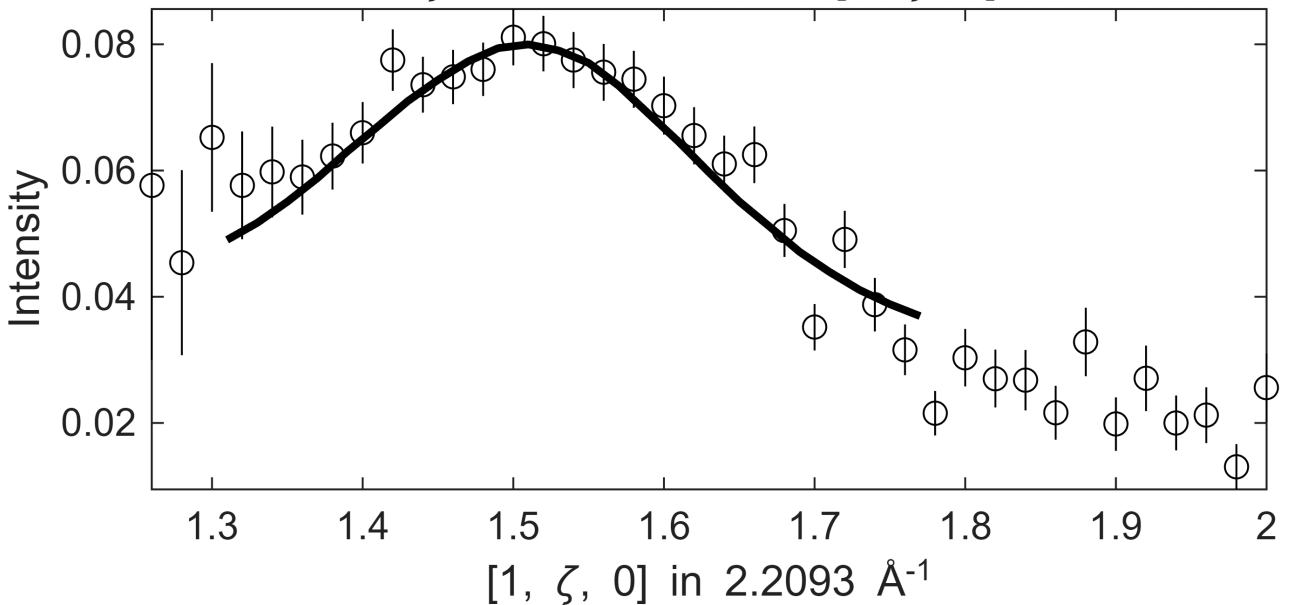
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 20#22 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.082919
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.69444

Fe_ei800, w2_800o11 Sym fits.sqw

S= 0.6 ± 0.071 ; J0= 35 ± 0.44 ; gamma= 111.649 ± 6.77886

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $260 \leq E \leq$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 260 \leq E \leq 270$$

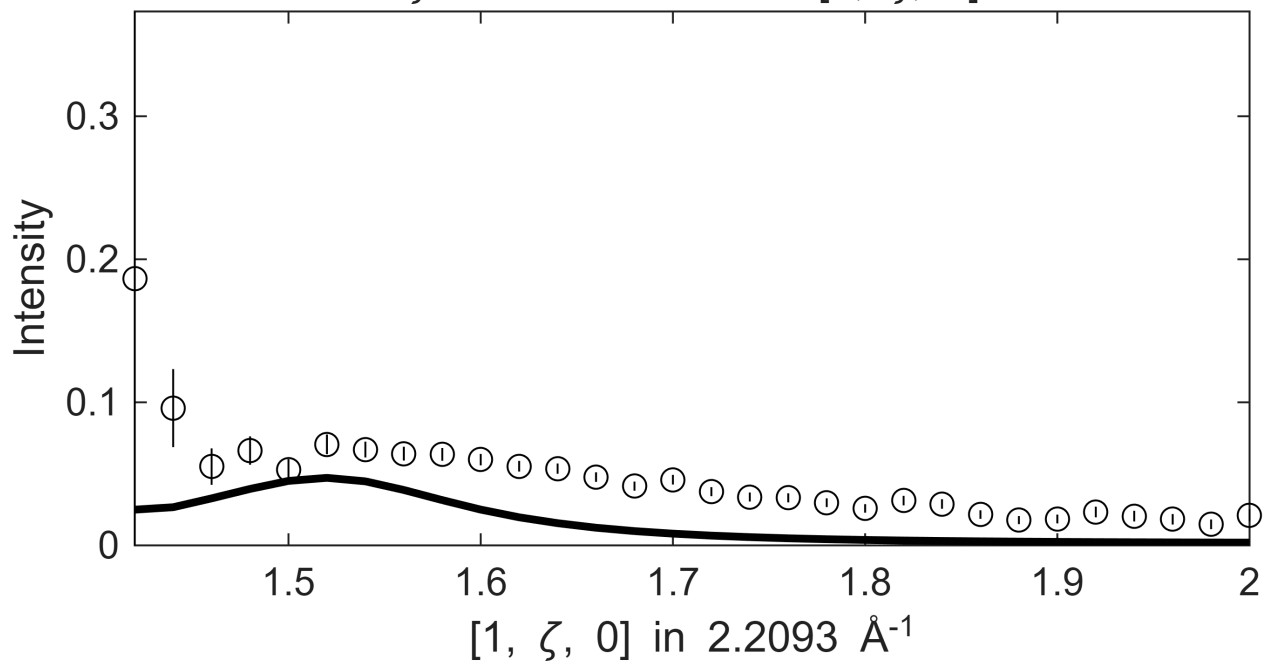
$$\zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


S=0.57± 0.2; J0= 34± 1.6; gamma=112.971±37.7167

$$-0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 270 \leq E \leq 280 \text{ in } [E, 0, 0],$$


$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 270 \leq E \leq 280 \text{ in } [E, 0, 0], \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$

$$.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 280 \leq E \leq 290, \zeta = -2.01:0.02:2.01 \text{ in } [1, \zeta, 0]$$


$$-0.1 \leq \xi \leq 0.1 \text{ in } [1-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [1, 0, \eta], 280 \leq E \leq 320 \text{ in } [E, 0, 0],$$
[illegible]

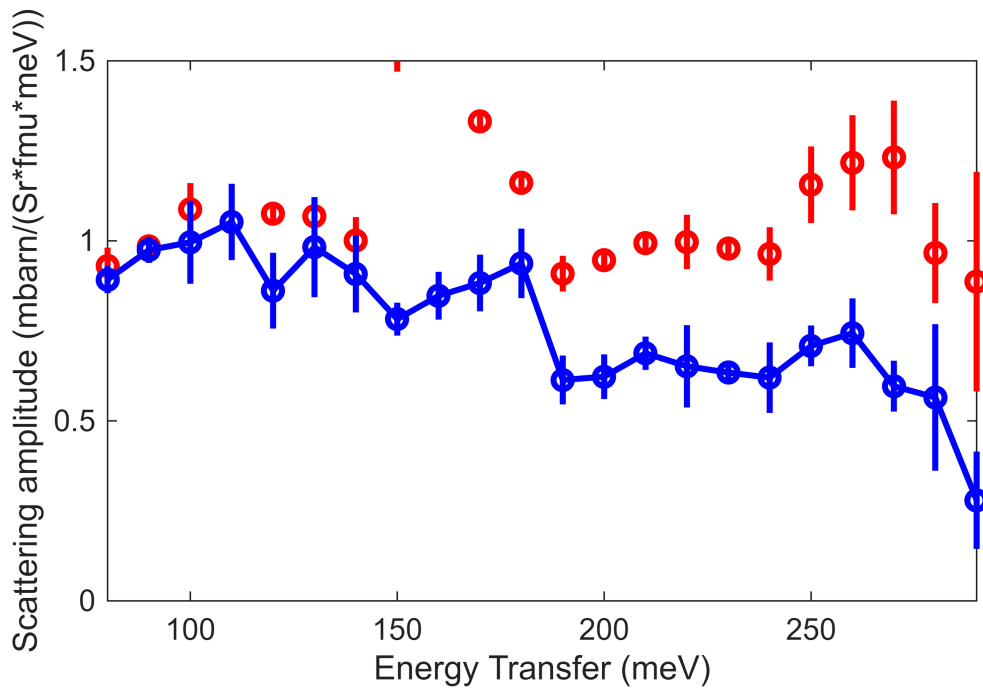
```

num_bp = cellfun(@(x)(numel(x.bp)),peak_fit_par);
mbp = max(num_bp);
bp_plot = nan(mbp,numel(num_bp));
bp_sig = nan(mbp,numel(num_bp));
en_axis = Sg.x;
for i=1:numel(peak_fit_par)
    bpl = peak_fit_par{i}.bp;
    bgs = peak_fit_par{i}.bsig;
    for j=1:numel(bpl)
        if iscell(bpl)
            bp_plot(j,i)= bpl{j}(1);
            bp_sig(j,i) = bgs{j}(1);
        else
            bp_plot(j,i) = bpl(1);
            bp_sig(j,i) = bgs(1);
        end
    end
end
bg_data = repmat(IX_dataset_1d(),1,4);
for j=1:4
    bg_data(j) = IX_dataset_1d(en_axis,bp_plot(j,:),bp_sig(j,:));
end
pd(bg_data); ly -10 10; keep figure;

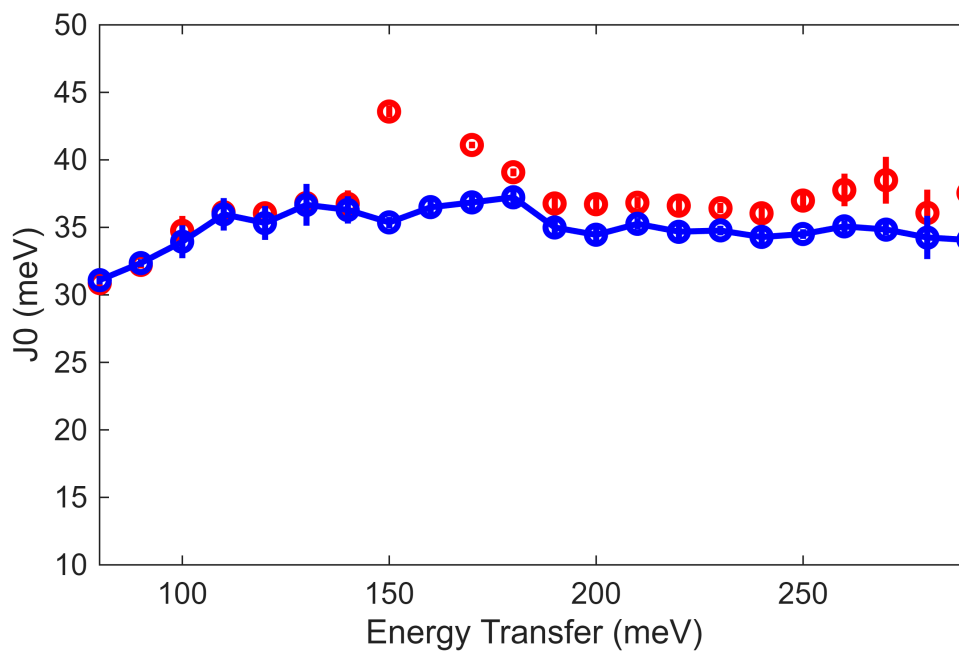
```

acolor **r**

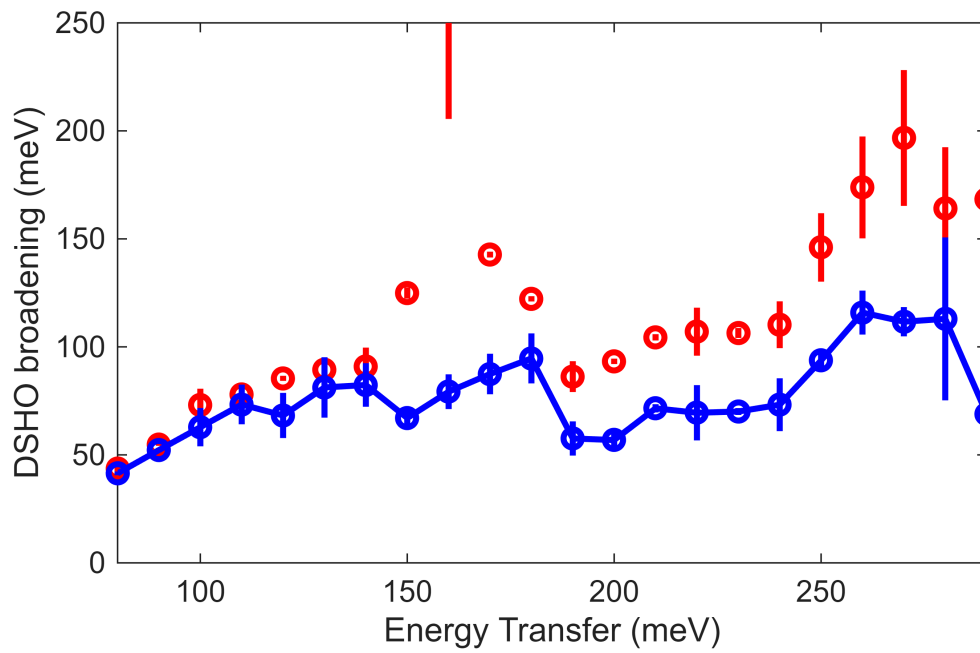
```
plot(fit_res.S);ly 0 1.5;liny;
acolor b
pd(Sg);ly 0 1.5;liny;keep_figure
```



```
acolor r
plot(fit_res.J0);ly 10 50;liny;
acolor b
pd(J0);ly 10 50;keep_figure
```



```
acolor r
plot(fit_res.gamma);liny;
acolor b
pd(G);ly 0 250;keep_figure
```



```
% fit_res.Speak_varsl = Sg;
% fit_res.J0peak_varsl = J0;
% fit_res.gamma_peak_varsl = G;
fit_res.Speak = Sg;
fit_res.J0peak = J0;
fit_res.gamma_peak = G;

fit_res.peak_fit_par = [peak_fit_par{:}];
fit_res.peak_fit_FixSlope_par = fit_res.peak_fit_par;
fit_resEi800 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi800");
```